

Project Atlas

Every experiment stage, its mathematics, its worked examples, and what each one confirmed - with all headline numbers recomputed from the project's own saved artifacts for this document

1. Legend: every symbol and every dimension

Symbol	Meaning	Shape / value
v	MobileCLIP image vector (phone)	[512], unit length
W	the adapter - ridge-fitted matrix, the deliverable	[512 x 768], 0.79 MB fp16
$u = v \cdot W$	predicted SigLIP vector, before normalization	[768]
$\hat{u}$	u after L2 normalization - what is stored	[768], unit length
y	TRUE vector: SigLIP's own embedding of the same image	[768], unit length
t	a text query through SigLIP's text tower	[768], unit length
X, Y	stacked training pairs (rows = images)	[3000 x 512], [3000 x 768]
A, B, R	pooled LLM activations: GPT-2, Pythia, random control	[13 x 10000 x 768] each
R-squared	variance explained, number-by-number; punishes length + direction errors	1 perfect, 0 = mean, negative = memorized-then-shrunk
cosine	angle-only similarity; what the index computes	$\hat{u} \cdot t$ after L2
CKA	coordinate-free similarity of two representation sets	[0, 1], invariant to rotation / permutation / uniform scale
R@K	fraction of queries whose true item ranks in top K	gallery size 1,000 here
13	hidden-state levels: embedding output (layer 0) + 12 blocks	both LLMs
768	width of GPT-2, Pythia, and SigLIP spaces	-
512	MobileCLIP output width - the governing dimension for the adapter fit	3000 / 512 = 5.9 rows per input dim
10,000 / 3,000 / 1,000	probe passages (Exp A) / training pairs / held-out images (Exp B)	-

2. The five models and the three spaces

Model	Role	Output space	Trained here?
GPT-2 (124M)	Exp A model A - frozen text encoder	13 levels x 768	no - frozen
Pythia-160M	Exp A model B - frozen text encoder, no shared ancestry with GPT-2	13 levels x 768	no - frozen
Random-weights Pythia	the control: same architecture, zero learning	13 levels x 768	no - by design
MobileCLIP-S1	phone-tier image encoder (Exp B source)	512	no - frozen
SigLIP 2 base	server-tier image + text encoder (Exp B target); its text tower makes every query	768	no - frozen

Three spaces matter. GPT-2's space and Pythia's space (Experiment A) are compared but never merged. SigLIP's space (Experiment B) is the **shared space**: the index lives in its coordinates, server photos' addresses are computed natively in it, phone photos' **addresses** are translated into it by W (the photo itself never moves and no description of it is ever created), and every query is written in it by SigLIP's text tower - which matches text to addresses on the semantic map its own training built, and never learns the adapter exists. The only fitted objects in the whole project are 13 stitching matrices, W , one Procrustes rotation, and one deliberately redundant MLP; every model ends the project byte-identical to how it was downloaded.

3. Experiment A - do two strangers share structure?

3.1 Stage A1: extraction

10,000 WikiText passages go through all three models. Each passage, at each of 13 levels, yields one 768-number vector by masked mean pooling over its tokens. Pooling is forced by alignment: the two tokenizers produce different token sequences, so token-to-token correspondence does not exist - passage-to-passage does. The loss is one-directional (blurring destroys distinctions, never invents shared ones), so every similarity below is a lower bound. Storage: $3 \times [13, 10000, 768] = 1.2$ GB.

3.2 Stage A2: CKA - comparing structure without coordinates

Weights cannot be compared (permutation symmetry: reorder one layer's neurons and the function is bit-identical while weight cosine drops to -0.125). CKA instead compares each model's internal similarity table over the same passages:

$$\text{CKA}(X, Y) = |Y^T X|_F^2 / (|X^T X|_F - |Y^T Y|_F)$$

Measured, all 169 layer pairs, against the control: trained mean 0.359 vs random 0.068 - a 5.3x contrast. The diagonal is elevated (0.425 vs 0.354, permutation $p < 0.001$) and **risks strictly monotonically from layer 3 to layer 11** (0.125 to 0.742, Spearman $\rho = 1.000$, $p < 10^{-5}$). The raw panel looks blocky because it is hubness-dominated - 97% of its energy is rank-1, row means span 0.19-0.57; with marginals divided out, 77% of layers match within two ($p = 0.006$). Full analysis: report Appendix C.7.

3.3 Stage A3: stitching - the strong test

Per layer pair, one ridge map from GPT-2's space to Pythia's, scored on held-out rows:

$$W = (X^T X + aI)^{-1} X^T Y$$

The inverse contains only X , so the 768x768 fit decomposes into 768 independent regressions sharing one inverse - the requirement is rows per input dimension. Run 1 had 1.6 (starved; control at R-squared -0.773, the memorize-then-shrink signature); run 2 has 9.8. Result: mean R-squared 0.737-0.741, peak 0.798 at layer 11, random control 0.173 - and the two curves have opposite slopes (trained rises 0.59 to 0.80, control decays 0.38 to 0.09), which no shared architecture can produce.

3.4 Direction symmetry and anisotropy (A3-bis / A3-ter)

Fitting the reverse direction looked asymmetric; two tests explained it completely. Layer 0 is a real information difference - sequence length is recoverable from GPT-2's pooled layer 0 at R-squared 1.000

(absolute positional embeddings) versus 0.126 for Pythia (rotary, applied inside attention). The middle-layer gap is a metric artifact: recomputed with uniform instead of variance weighting it collapses from 0.097 to 0.006. The cause is measured anisotropy - effective dimensionality 2.2-10.9 (GPT-2) versus 9.3-92.7 (Pythia). With the artifact removed the isomorphism is symmetric, as the hypothesis predicts.

4. Experiment B - the finding becomes a product

4.1 Stages B1-B4 in one flow

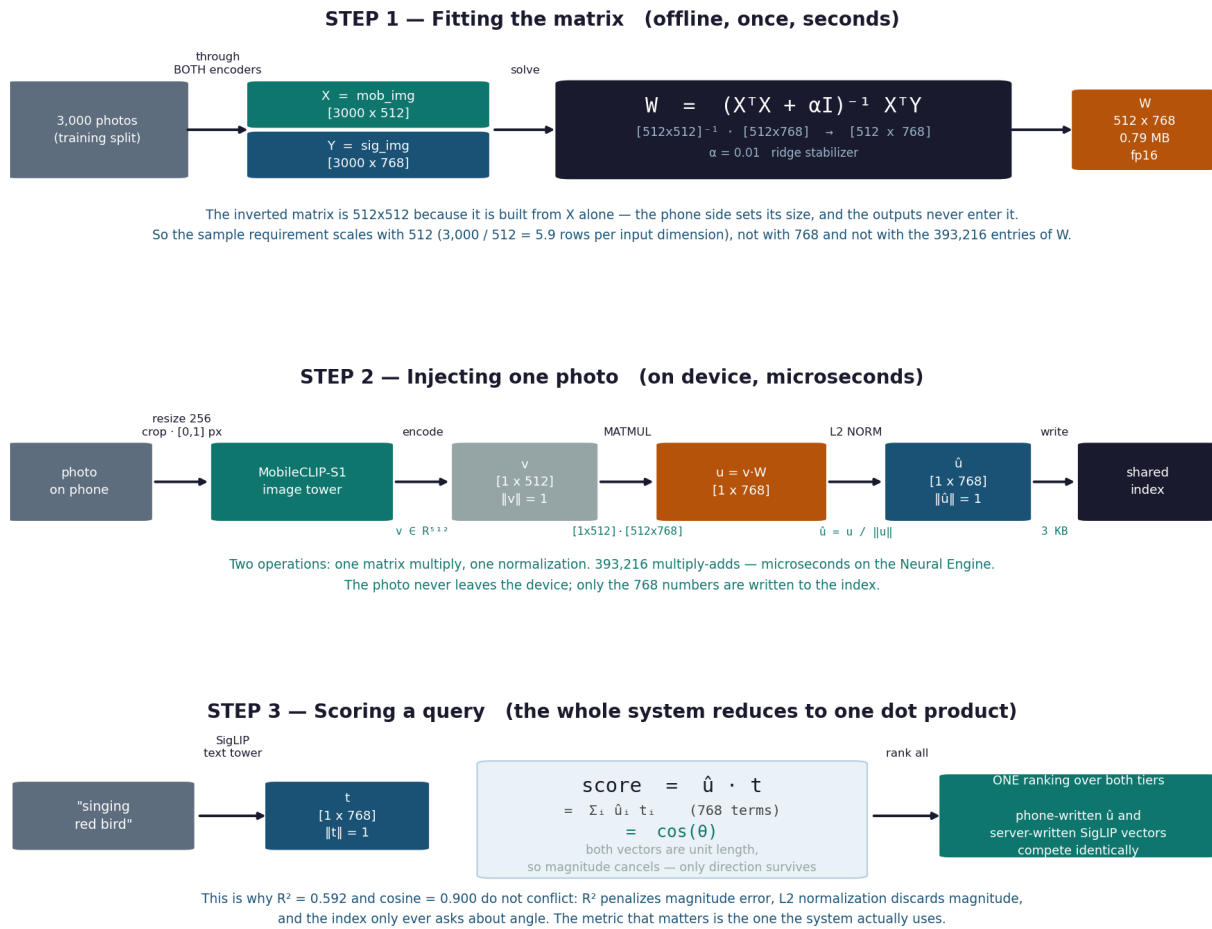


Figure 1: fit W offline from 3,000 image pairs (the inverted matrix is 512x512 - built from the phone side alone); inject each photo with one matmul + one normalization (393,216 multiply-adds, microseconds on the Neural Engine); score every query as one dot product, which after L2 normalization is exactly cos(theta).

4.2 The numbers - recomputed from the saved artifacts for this document

Gallery (1,000 held-out images)	R@1	R@5	R@10
A - SigLIP native (ceiling)	0.630	0.885	0.944
B - MobileCLIP + W (the system)	0.557	0.842	0.918
B as % of ceiling	88.4%	95.1%	97.2%
C - raw MobileCLIP zero-padded	0.001	0.005	0.010

Mean cosine to target: adapted 0.900, raw -0.003 (exact chance in retrieval), Procrustes rotation 0.775 direction but R-squared 0.038 - a rotation cannot fix anisotropy mismatch, the same property measured in Experiment A. The MLP control (512-1024-768, 300 epochs, the only gradient training in the project) improved cosine by +0.003: the relationship is linear, and the residual gap is missing information from MobileCLIP's 1024-to-512 bottleneck, not translation error. The fp16 export is bit-exact against the fp32 matrix cast to half precision (max difference 0.0) and reproduces cosine 0.900.

4.3 The geometry, from the real held-out embeddings

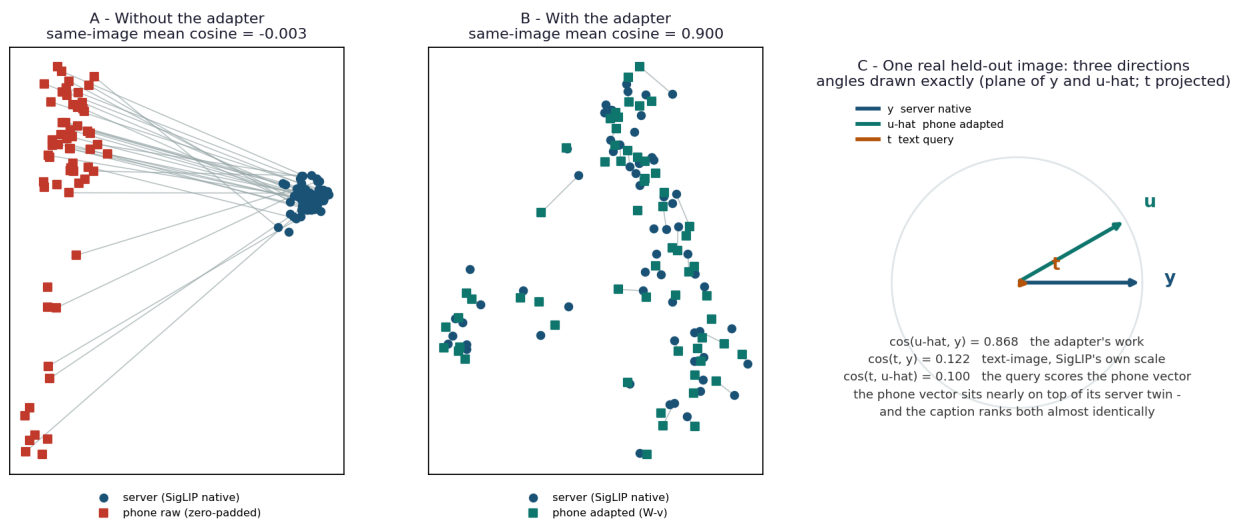


Figure 2: real evaluation vectors, one shared 2-D PCA per panel. Without W the two encoders occupy disjoint regions (mean same-image cosine -0.003); with W each phone vector lands beside its server twin (0.900). Panel C draws one actual image's three directions with exact angles: $\cos(u\text{-hat}, y) = 0.868$ for this image, while the caption sits at $\cos(t, y) = 0.122$ and $\cos(t, u\text{-hat}) = 0.100$ - SigLIP's sigmoid training keeps text-image cosines low (diagonal mean 0.150) while ranking stays sharp, and the query scores the phone vector and the server vector almost identically, which is the entire product claim in one picture.

4.4 One worked example, fixed throughout the project

Anchor $y = (0.60, 0.80)$, length 1. Chapter 1: $u = 0.85y = (0.51, 0.68)$ - cosine $0.850/0.850 = 1.000$, squared error $0.09^2 + 0.12^2 = 0.0225$: R-squared punished, cosine perfect, normalization erases the error. Chapter 2: keep $|u| = 0.85$, rotate to $u = (0.75, 0.40)$ - cosine $0.770/0.850 = 0.906$ (about 25 degrees), squared error $0.1825 = 0.1794$ angular (98.3%) + 0.0031 magnitude (1.7%): normalization removes only the 1.7%. Between chapters only direction changed and only cosine moved - cosine measures direction alone. Real system: R-squared 0.592 (length + direction, pre-normalization) vs cosine 0.900 (direction only, post) - and the index normalizes first, so cosine is the product metric.

4.5 How the shared space actually works: address, map, and volume

No description is ever created. The phone photo is never captioned, tagged, or described; no text representation of it exists anywhere in the system. What crosses the tier boundary is an **address**: MobileCLIP's judgment of where this photo belongs in semantic space, re-expressed by W in SigLIP's coordinates. Text-to-image matching was solved once, inside SigLIP, during its own contrastive training - which arranged one 768-d space so that a photo and a text describing the same thing point in the same direction. The adapter does not re-solve matching; it grants MobileCLIP's photos access to that already-solved matching by translating their addresses into the space where the solution lives. Nearness to the right text is **inherited from the space, not assigned**: the index write computes $u\text{-hat} = L2(v-W)$ with no knowledge of any query, and if the address is correct it lands in the region the matching text already points at. Measured on one real held-out image (Figure 2, panel C): the caption scores the server twin at cosine 0.122 and the adapted phone vector at 0.100 - nearly identically, and the text tower never learns the adapter exists.

Map, coordinates, geography. Two cartographers who never met survey the same country: no city shares an (x, y) position across their maps (different north, units, origin), yet the relative layout - which cities are near which - is identical, because both surveyed the same country. Consequently one fixed transformation converts every position on one map to the other. Dictionary: the country is visual semantics; a map is an encoder's space; coordinates are the arbitrary basis a training run settled into; geography is the similarity structure; the one transformation is W . This is what makes the experiment falsifiable: a linear map can change coordinates but cannot move one city relative to another. Had the two encoders disagreed about geography - not merely coordinates - no W could help, and retrieval would sit at the raw baseline (0.001, exact chance). It reaches 95% of ceiling instead: the geographies already matched, and W merely re-labelled the axes.

What W does - and the volume conversion that carries most of the work. By the SVD, $W = U-S-V^T$: rotate, scale each direction by its own singular value, rotate again. Rotation converts *which direction means what*; the diagonal S converts *how loudly each meaning is spoken*. The two encoders disagree about both. Toy with exact numbers: suppose MobileCLIP writes indoor/outdoor at +5 and day/night at +0.2, while SigLIP writes them at +1 and +3. An outdoor-day photo is (5, 0.2) and must become (1, 3). With axes already aligned, a rotation is finished - and delivers (5, 0.2), wrong by 5x and 15x, on every photo, because rotations preserve all lengths by definition. The fix is $S = \text{diag}(0.2, 15)$: $5 \times 0.2 = 1$, $0.2 \times 15 = 3$ - one fixed pair of volume knobs correcting the whole population, since loudness is a property of the axis, not of individual photos. Ridge finds these knobs automatically: $X^T X$ carries the source loudness per direction, $X^T Y$ the target loudness, and their combination computes the ratios from the 3,000 pairs.

Measured on the shipped W. Its singular values span 0.052 to 3.592 - a 69x spread, with 81% of the 512 knobs outside the neutral band [0.8, 1.25]. A rotation is the same machine with every knob welded to 1, which is why Procrustes collapsed: its output points substantially the right way (cosine to target 0.775) while scoring R-squared 0.038 - right meanings, catastrophically wrong volumes. That pair of numbers is the anisotropy diagnosis in miniature, and the 0.038-to-0.592 gap is the measured size of the volume-conversion problem. What W cannot do is equally precise: nothing per-image - it cannot move one photo's address relative to another's or add detail MobileCLIP did not encode (the MLP's +0.003 proves nothing beyond the rigid transformation was needed). The adapter is grammar between two dialects: it edits how the address is written, never what the address says.

5. Experiment C - how far without paired data? (B10 / C1)

The adapter needed 3,000 paired images. vec2vec-style translation asks whether the correspondence is recoverable with **zero** pairs, from distribution shape alone. Built: Gromov-Wasserstein initialization ($n = 800$, $\epsilon = 3e-3$) + CSLS mutual-neighbour refinement, on disjoint image sets. Result: supervised reference cosine 0.835 / R@1 0.910; unsupervised cosine 0.395 - 23x above the 0.017 random floor, so distribution-level alignment is real - but instance-level R@1 = 0.002, exact chance, frozen at 428 mutual pairs. Diagnostics refuted the calibration-mismatch hypothesis and supported continuous-manifold + anisotropy mismatch: image manifolds lack the discrete anchor structure word-embedding spaces offer. Conclusion recorded honestly: shape alone recovers the coarse map, not the fine one; the full adversarial vec2vec machinery is necessary, not decorative. (Report Appendix B.)

6. Experiment D - guarding the measurements, and a bonus capability

6.1 Power diagnostics (D1)

Before believing any negative result, three checks in triage order: rows per input dimension (starvation biases downward, systematically - run 1's lesson); the shuffle test (fit on deliberately misaligned rows and require the honest fit to win by a wide margin - catches the silent off-by-one that no error message reports); artifact sanity (NaN / zero rows / duplicates / unit norms). Validated: aligned synthetic data scores 1.0 against -0.017 shuffled.

6.2 Cross-tier membership - predicted, now measured

Appendix D predicted that querying the shared index with the adapted phone vector itself (image-as-query) would answer 'is this photo already in the server library?'. Measured for this document on the 1,000 held-out images:

Image-as-query, adapted -> SigLIP gallery	value
R@1 (true twin ranks first)	0.987
R@5 / R@10	0.999 / 0.999
membership gap (true-twin cosine minus best impostor), mean	+0.075
fraction of images with positive gap	98.7%

Text-to-image runs at 88-97% of ceiling because captions are ambiguous; image-to-image is near-perfect because the query and the target encode the same pixels with no modality gap. A simple threshold on the gap implements cross-tier deduplication without uploading the photo - a fourth product capability the adapter creates (raw vectors score chance).

7. What the project confirmed

Claim	Evidence	Where
Independently trained LLMs share structure recoverable by one linear map	held-out R-squared 0.737-0.741 mean, 0.798 peak; control 0.173; opposite depth slopes	A3
The correspondence deepens with depth	CKA diagonal strictly monotone L3-L11 ($\rho = 1.000$) AND stitching rises over the same span - two independent measures	A2 + A3
The correspondence is symmetric	reverse-direction gap 0.097 to 0.006 under uniform weighting; layer-0 exception is architectural and measured	A3-bis/ter
The relationship is linear, not merely learnable	MLP with 10x parameters and direct cosine training gains +0.003	B2-bis
Rotation alone is insufficient - the spaces differ in anisotropy	Procrustes R-squared 0.038 vs ridge 0.592; eff. dim 2.2-10.9 vs 9.3-92.7 measured	B2 + A3-ter
The finding ships: two encoders, one index	R@5 95.1% of ceiling; 0.79 MB fp16 artifact, bit-exact export; 39 ms end-to-end demo	B3-B6
Cross-tier membership works	image-as-query R@1 0.987, positive gap 98.7%	B3-ter
Zero pairs recover coarse but not fine alignment	unsupervised cosine 23x floor yet R@1 chance; diagnosis documented	B10/C1
Negative results require power checks first	run 1: 1.6 rows/dim, control -0.773; rerun with 9.8 flipped the verdict, nothing else changed	App. C

Scope, stated plainly: two small language models and one vision pair are point samples of the Platonic Representation Hypothesis, consistent with its trend and not proof of it.

8. Where to go forward

Wider bottleneck. The R@1 gap is concentrated in MobileCLIP's single 1024-to-512 projection; refitting W against MobileCLIP-B or MobileCLIP2 (wider output) is a seconds-cheap test of the diagnosis. **Same-dimensional Procrustes.** The 0.038 mixes anisotropy with a zero-padding handicap; two 768-d encoders separate the causes. **More points on the curve.** Repeat Experiment A across model scale to test the PRH trend rather than one point of it. **Server-side re-rank.** The R@1 escape hatch (re-rank top-10 natively) is designed but unmeasured. **Hebrew editions.** Frozen pre-appendix; refresh after the defense. **Live index.** The B6 demo is 9 vectors; the same code against a production Qdrant collection is the next engineering step.